LESSON PLAN · GENERATIVE AI (GR 6–8) INTERMEDIATE: GEN-AI CREATORS · LEVEL 2

Lesson 8: Capstone: AI for Our School

Module 2 | 60 minutes | Problem → prototype → pitch | Grades 7–11 · ages 12–16

Objective: Design, prototype, test and pitch an AI-powered helper that solves a real problem at your school, with an ethics check and an honest AI log.

Success criteria

- I can define a real problem and user.
- I can design and prototype an AI helper with rules and a human role.
- I can test my prototype and run an ethics check.
- I can pitch my solution in 3 minutes with a live demo.

Key vocabulary

- **capstone**: A final project that uses everything from the course.
- **prototype**: A small, working first version of an idea.
- **user**: The person the solution is designed for.
- **ethics check**: Checking a design for privacy, fairness, honesty and over-trust.
- **over-trust**: Relying on AI more than you should.
- **pitch**: A short, persuasive presentation of your idea.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Printed project brief
- Project files: l08-capstone-project-kit.md, l08-teacher-demo-script.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	Problem hunt: teams list school problems and vote for the most real, small and helpful one.
4 min	**Watch + poll**	Play the capstone video. Poll: which kind of prototype will you build — app, program, chart or video?
5 min	**Live demo**	Teacher shows the model Recycling Guide: tests, a caught mistake, the ethics prompt and the pitch timer.
5 min	**Design (pairs)**	Pairs write problem, user and spec, then the critical-friend Prompt Lab; improve until ticked.
18 min	**Build & test**	Build the prototype and run 6 tests with fixes. Halfway thumbs check: is something working? Ethics sort for anyone waiting for help.

4 min	**Quick competition**	Pitch order race — first three correct choose the pitch order.
5 min	**Breakout: pitch practice**	In breakout rooms, practise the pitch to a 3-minute timer; partners ask the judge's question.
10 min	**Show-and-tell pitches**	Volunteers pitch live with a demo (the rest submit recorded or written pitches). Class gives two stars and a wish in the chat.
4 min	**Probe + submit**	New situation: the judge's question, then submit the capstone and reflection.

Unplugged warm-up

Problem hunt (5 min): learners list problems they noticed at school this week (lost items, queues, recycling, revision). Each team picks their top three and votes: which is real, small and helps someone? Discuss which ones AI could actually help with — and which need a human instead.

Teaching the idea

The capstone pulls together Level 2: coding with an AI partner (Module 1) and data, media and no-code apps (Module 2). Each learner or pair picks a real, small school problem, names the user, writes an AI spec (inputs, outputs, rules, the human's role), builds a prototype (app instructions, a short program, a survey chart or a video plan), tests it with normal, tricky and attack cases, runs an ethics check (privacy, fairness, honesty, over-trust) and gives a 3-minute pitch with a live demo. AI may act as a critical friend, explainer and debugger; the idea, build and pitch are the learner's, with a what-AI-did log. Keep the age rules: under 13 use the Kitabash AI helper; 13–17 approved tools with parent permission; Claude never.

Analogy: The capstone is like entering a science fair: the judges want to see a real problem, a working model, honest test results and that you know its limits — not a perfect poster.

Live demo: Show a model capstone in 5 minutes: the Recycling Guide. 1) Problem slide: "Students are unsure which bin to use." 2) Paste the guide's instructions into the Kitabash AI helper and run a normal test ("banana peel") and a tricky test ("used chips packet"). If it says paper, "Caught it!" — add the school bin list and an "ask a teacher if unsure" rule, and rerun. If it answers well, try "a broken pen" and show it admitting uncertainty. 3) Type the critical-friend ethics prompt and pick two risks to address. 4) Show the pitch structure with a 3-minute timer.

Questions to ask

Ask	Listen for
What makes a good capstone problem?	Real, small enough to prototype, and helps a named user.
Name the four parts of the ethics check.	Privacy, fairness, honesty, over-trust.
What should your helper do when it is unsure?	Say so and point to a human, like a teacher or the office.

Why include failed tests in your pitch?	They show you tested honestly and improved the design.
What goes in an honest AI credit?	What AI helped with (e.g. ethics ideas, debugging) and that you checked it.

Differentiation

Support
Offer ready templates: the Revision Buddy app or the survey report as a base. Pitches can be 2 minutes with a partner sharing the talking.

Stretch
Add a short Python or Scratch component (for example a recycling quiz) or a survey of real users (anonymous) with a chart showing whether they would use it.

Common misconceptions

Misconception	How to fix it
A capstone must use AI for everything	Use AI where it helps. Choosing a human step on purpose is good design.
A great pitch hides the problems	Showing what failed and how you fixed it builds trust.
If the AI said the ethics risks, the check is done	Check and choose the risks yourself; AI may miss local ones or invent others.
The prototype has to be finished and perfect	A small working version that is tested is exactly what a prototype is.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Problem & user	Problem unclear or not real	Problem stated, user vague	Real problem and clear user, checked with people	Sharp, well-evidenced problem with a clear user need
AI design & prototype	No working prototype	Prototype partly works; spec incomplete	Working prototype with a clear spec and human role	Thoughtful design: AI used only where it helps, strong rules, polished prototype
Testing & ethics	No tests or ethics check	A few normal tests; ethics check vague	6+ tests with fixes; ethics check covers all four areas	Attack and edge cases tested and fixed; ethics risks addressed in the design

Pitch & honesty	No pitch or no AI credit	Pitch over time or missing a demo	Clear 3-minute pitch with demo and honest AI credit	Compelling pitch, smooth demo, confident answers about limits

Exit ticket

What is one thing your AI helper must never be trusted to do, and how does it tell users?

Homework

Pitch your capstone to your family. Ask them the judge's question and one of their own. Write down their feedback and one improvement for version 2.

Answer key — Kitabash activities

1. Submit: AI for our school (probe · project)

2. Prompt Lab: critical-friend ethics check (perform · ai lab)

Checklist: Describe the problem and user · Ask it to be critical · Privacy and fairness · Honesty and over-trust · Ask for fixes · Keep ownership

Strong example:

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My idea is a recycling guide for students at my school: they type an item and it tells them which bin to use. Act as a critical friend. List the risks under privacy, fairness, honesty and over-trust, and suggest one fix for each. Do not rewrite my idea — I will decide what to change.
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3. Ethics check: pass or needs a fix? (perform · categorize)

Passes: The helper says “Ask at the school office” when unsure, It has a note: “AI can be wrong — check with a teacher”, Test log shows attack tests were run and fixed
Needs a fix: It asks students for their full name and class, It only works for students who read English well, It invents where a lost bottle is

4. Pitch order (perform · ordering)

Order: The problem, with a story or checked number → Who the user is → What your AI helper does → A live demo with a normal and a tricky test → What you tested, fixed and checked for ethics → Next step and an honest AI credit

5. New situation: the judge’s question (probe · multiple choice)

Answer: “It only suggests places to check, says when it does not know, and always tells students to ask at the office. We tested it with tricky cases.”

Answer key — worksheet

1. **Key words:** capstone = A final project that uses everything from the course. · prototype = A small, working first version of an idea. · user = The person the solution is designed for. · ethics check = Checking a design for privacy, fairness, honesty and over-trust.
2. **Prompt Lab: critical-friend ethics check:** Include: Describe the problem and user; Ask it to be critical; Privacy and fairness; Honesty and over-trust; Ask for fixes; Keep ownership. Example: My idea is a recycling guide for students at my school: they type an item and it tells them which bin to use. Act as a critical friend. List the risks under privacy, fairness, honesty and over-trust, and suggest one fix for each. Do not rewrite my idea — I will decide what to change.

3. **Ethics check: pass or needs a fix?:** Passes: The helper says “Ask at the school office” when unsure, It has a note: “AI can be wrong — check with a teacher”, Test log shows attack tests were run and fixed · Needs a fix: It asks students for their full name and class, It only works for students who read English well, It invents where a lost bottle is
4. **Pitch order:** 1 The problem, with a story or checked number · 2 Who the user is · 3 What your AI helper does · 4 A live demo with a normal and a tricky test · 5 What you tested, fixed and checked for ethics · 6 Next step and an honest AI credit
5. **New situation: the judge’s question:** B. “It only suggests places to check, says when it does not know, and always tells students to ask at the office. We tested it with tricky cases.” — Strong designers expect mistakes, design for them (suggest, admit uncertainty, hand over to a human) and show the tests that prove it.
6. **My plan:** A clear plan that covers every requirement.
7. **Checklist:** All requirements met.
8. **Two stars and a wish:** Kind, specific feedback and an improvement.

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